

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, III: THE IWAHORI-HECKE TOWER AND THE TYPE III DISSOLUTION OF THE TRACE OBSTRUCTION

RUQING CHEN

ABSTRACT. Papers I–II established that the arithmetic of the GL_2 Hecke–Laplacian $\Delta_p = (q+1)I - T_p$ — the critical group $\mathrm{Jac}(Y)$ of order $\frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$ — is carried by a virtual KK -class whose companion matrix C satisfies $\mathrm{Tr}(C^2) = -n(q-1) < 0$, so that no non-negative C^* -correspondence of any size realizes it (the unconditional trace obstruction, [II, Thm. 1.6]). This paper takes the projective limit over the tower and determines what survives.

Section 1 proves that the geometric tail equivalence relation on the Cantor limit $\Omega = \varprojlim_k V_{Y_k}$, with its harmonic measure ν , has Radon–Nikodym cocycle valued in $q^{\mathbb{Z}}$ and asymptotic ratio set $r_\infty = q^{\mathbb{Z}} \cup \{0\}$, computed in full by a density argument; by Krieger’s theorem the associated von Neumann algebra is the hyperfinite factor of type $\mathrm{III}_{1/q}$, the Powers factor $R_{1/q}$, and for Satake-twisted cocycle classes with range $\mu_0^{\mathbb{Z}}$ the same computation gives type $\mathrm{III}_{\mu_0 \wedge \mu_0^{-1}}$. A type III factor carries no faithful normal semifinite trace, so the inequality $\mathrm{Tr}(\tilde{C}^2) \geq 0$ that powered the finite No-Go theorems has no formulation on the limit: the obstruction dissolves as a *sentence*, not as a defeated inequality. We prove equally that the dissolution buys less than it appears to: any inductive-limit realization by finite-level correspondences still dies level by level, and the weak closure that dissolves the trace also annihilates the K -theory the programme wants to realize — the correct target is a traceless Kirchberg C^* -limit, not a von Neumann algebra, and we say so.

Section 2 formalizes the finding of [II, Rem. 5.2]: the congruence (Iwahori path) tower obeys the exact pseudo-Iwasawa law $\nu_2 = 6 \cdot 2^k - k - 4$ with $\lambda = -1$, impossible for a $\mathbb{Z}_p[[T]]$ -characteristic ideal. The one-step recursion $|\mathrm{sandpile}(Y_{k+1})| = q^{n_k(q-1)-1} |\mathrm{sandpile}(Y_k)|$ — whose exponent corrects the $q = 2$ special form of earlier drafts — is proved here outright, self-containedly, from the AB/BA spectral identity for line digraphs together with the equal-cofactor eigenvalue-product formula for Eulerian Laplacians, making Section 2 unconditional; the full law follows in two lines, and $\lambda = -1$ is identified as $-\chi$ for the one-dimensional Eisenstein kernel removed at each level of the degeneracy correspondences: an Euler-characteristic correction over the Iwahori–Hecke base, external to any commutative Iwasawa module. A closing outlook section states, without derivation, the p -adic horizon.

STATUS OF THE RESULTS

Theorem 1.5 and its supporting lemmas are proved here in full up to two classical inputs, quoted and flagged: Krieger’s ratio-set criterion for the type of an ergodic amenable relation, and the uniqueness of the hyperfinite type III_λ factor for $0 < \lambda < 1$. Corollary 1.7 and Proposition 1.8 are complete. Section 2 is now unconditional: the line-digraph recursion, previously a five-level computational hypothesis, is proved in full as Theorem 2.1, self-containedly and with the exponent corrected from $n_k - 1$ (a $q = 2$ artifact) to $n_k(q-1) - 1$; the computational confirmations ($q = 2$,

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Paper III of the series. Papers I–II are cited internally as [I] and [II]; their numbered results are quoted with those prefixes. External citations are subject to the series’ standing caveat — bibliographic data from recollection and out-of-band checks — except where an entry is explicitly marked as verified against its primary source.

five levels; $q = 3$, one instance) now confirm a theorem rather than support a hypothesis. Section 3 derives nothing beyond one exactly verified bookkeeping identity, clearly labelled.

1. THE TYPE OF THE LIMIT, AND WHAT DISSOLVES

1.1. The Cantor limit and its harmonic measure. Throughout, $q \geq 2$. Let X be the $(q + 1)$ -regular tree with base vertex x_0 and let

$$\Omega = \{1, \dots, q + 1\} \times \{1, \dots, q\}^{\mathbb{N}}$$

be its geodesic boundary in rooted coordinates: the first symbol chooses among the $q + 1$ edges at x_0 , every later symbol among the q non-backtracking continuations. Under the identifications of [I, §1], $\Omega \cong \partial X \cong \varprojlim_k V_{Y_k}$ for the Iwahori path tower, a Cantor set. The *harmonic measure* ν is the product measure with uniform factors: mass $\frac{1}{q+1}$ on the first coordinate, $\frac{1}{q}$ on each subsequent one, so that every cylinder $[w]$ of length $n \geq 1$ has $\nu([w]) = \frac{1}{q+1} q^{-(n-1)}$.

Definition 1.1 (Tail relation and degree cocycle). $\mathcal{R}$ is the tail equivalence relation on Ω : $\omega \sim \omega'$ iff $\sigma^m \omega = \sigma^n \omega'$ for some $m, n \geq 0$, where σ deletes the first symbol. The *degree cocycle* of the pair (ω, ω') realized at (m, n) is $c = m - n \in \mathbb{Z}$; it is well defined on $\mathcal{R}$. $\mathcal{R}_0 \subset \mathcal{R}$ is the degree-zero subrelation.

Lemma 1.2 (Radon–Nikodym cocycle). ν is quasi-invariant under $\mathcal{R}$, and every partial isomorphism $\gamma \in [[\mathcal{R}]]$ of degree $c(\gamma)$ has

$$\frac{d\nu \circ \gamma}{d\nu} = q^{c(\gamma)} \quad \nu\text{-a.e. on its domain.}$$

In particular the cocycle takes values in the discrete group $q^{\mathbb{Z}} \subset \mathbb{R}_{>0}$, and $\mathcal{R}_0$ preserves ν .

Proof. It suffices to check the generating partial isomorphisms $T_{u,v}: [u] \rightarrow [v]$, $(u\xi) \mapsto (v\xi)$, for finite words u, v of lengths $m, n \geq 1$: these have degree $m - n$ and scale every cylinder inside $[u]$ by $\nu([v])/\nu([u]) = q^{m-n}$. Every element of $[[\mathcal{R}]]$ is a countable disjoint union of restrictions of such maps. $\square$

Lemma 1.3 (Ergodicity). $\mathcal{R}_0$, hence $\mathcal{R}$, is ergodic on (Ω, ν) .

Proof. A Borel set invariant under $\mathcal{R}_0$ is invariant under every change of finitely many initial coordinates (compose $T_{u,v}$ with $|u| = |v|$), hence is a tail event for the product measure ν ; Kolmogorov's zero–one law gives measure 0 or 1. $\square$

1.2. The ratio set. Recall Krieger's asymptotic ratio set $r_\infty(\mathcal{R}, \nu) \subseteq [0, \infty)$: $\lambda \in r_\infty$ iff for every $\varepsilon > 0$ and every Borel A with $\nu(A) > 0$ there is $\gamma \in [[\mathcal{R}]]$ with domain $B \subseteq A$ of positive measure, $\gamma(B) \subseteq A$, and $|d\nu \circ \gamma / d\nu - \lambda| < \varepsilon$ on B .

Theorem 1.4 (Ratio set of the geometric tail relation). $r_\infty(\mathcal{R}, \nu) = q^{\mathbb{Z}} \cup \{0\}$.

Proof. Upper bound. By Lemma 1.2 the derivative of every partial isomorphism is ν -a.e. $q^{\mathbb{Z}}$ -valued, and $q^{\mathbb{Z}}$ is closed and discrete in $\mathbb{R}_{>0}$; hence $r_\infty \cap \mathbb{R}_{>0} \subseteq q^{\mathbb{Z}}$.

Lower bound: $q \in r_\infty$. Let A have positive measure and let $0 < \delta < \frac{1}{3}$. By the martingale (Lebesgue density) theorem for the cylinder filtration, there is a point of A at which the cylinder densities converge to 1; choose nested cylinders $[u] \supset [us]$ (with $|u| \geq 1$, s the next symbol of that point) satisfying

$$\nu(A \cap [u]) > (1 - \delta) \nu([u]), \quad \nu(A \cap [us]) > (1 - \delta) \nu([us]).$$

Let $T = T_{us,u}: [us] \rightarrow [u]$, the degree- $(+1)$ tail map, with constant derivative q by Lemma 1.2. The bad set inside the domain is

$$[us] \setminus A \quad \text{of measure} < \delta \nu([us]), \quad T^{-1}([u] \setminus A) \quad \text{of measure} = \frac{1}{q} \nu([u] \setminus A) < \delta \nu([us]),$$

so $B := (A \cap [us]) \cap T^{-1}(A \cap [u])$ has $\nu(B) > (1 - 2\delta)\nu([us]) > 0$, and $T|_B$ maps a positive-measure subset of A into A with derivative exactly q . Hence $q \in r_\infty$; applying the same argument to T^{-1} and to compositions gives $q^{\mathbb{Z}} \subseteq r_\infty$.

The value 0. Since $r_\infty \cap \mathbb{R}_{>0} = q^{\mathbb{Z}} \neq \{1\}$, the relation is not of type II, and for an ergodic relation this forces $0 \in r_\infty$ by Krieger's dichotomy; alternatively, iterating $T_{us^k, u}$ realizes derivatives $q^k \rightarrow \infty$ on shrinking sets, whose inverses realize derivatives $\rightarrow 0$. $\square$

1.3. The factor and its type.

Theorem 1.5 (Type of the limit). *Let $M_\infty := W^*(\mathcal{R}, \nu)$ be the Feldman–Moore von Neumann algebra of the geometric tail relation. Then M_∞ is the amenable (hyperfinite) factor of type $\text{III}_{1/q}$, isomorphic to the Powers factor $R_{1/q}$. Moreover, for any quasi-invariant measure class on Ω whose Radon–Nikodym cocycle on $\mathcal{R}$ takes values in a subgroup $\mu_0^{\mathbb{Z}} \subset \mathbb{R}_{>0}$ with $\mu_0 \neq 1$ and for which the density argument of Theorem 1.4 applies verbatim — in particular for the Satake-twisted cocycle classes of [I, §2] whose weights range in $q^{-\beta\mathbb{Z}}$ — the factor is of type III_λ with $\lambda = \min(\mu_0, \mu_0^{-1})$; for the twisted classes, $\text{III}_{q^{-\beta}}$.*

Proof. $\mathcal{R}$ is a countable Borel equivalence relation with countable classes, generated by the maps $T_{u,v}$, hence amenable as an increasing union of finite subrelations (the level- N truncations of [I, Def. 1.6]); ν is quasi-invariant (Lemma 1.2) and ergodic (Lemma 1.3), so M_∞ is an amenable factor by Feldman–Moore. Krieger's theorem identifies the type of an ergodic amenable relation from its ratio set: $r_\infty \cap \mathbb{R}_{>0} = \lambda^{\mathbb{Z}}$ with $0 < \lambda < 1$ gives type III_λ . Theorem 1.4 gives $\lambda = 1/q$. Uniqueness of the amenable type III_λ factor for $0 < \lambda < 1$ (Connes) identifies M_∞ with the Araki–Woods/Powers factor $R_{1/q}$. The twisted statement repeats the proof with q replaced by μ_0 wherever the cocycle value enters; nothing else in the argument refers to the numerical value. $\square$

Remark 1.6 (A correction to the loose phrasing of [I, Thm. 2.4(iii)]). Paper I stated a family “type $\text{III}_{q^{-\beta}}$ ” for the boundary system. For the *untwisted* geometric cocycle this is misleading as a family: the uniform measure is the unique conformal probability class for every β (all branch weights being equal, the normalized cylinder masses are β -independent), the KMS point on the boundary quotient is unique, and the factor type is $\text{III}_{1/q}$, full stop. A genuine β -family of types arises only when the cocycle itself is deformed — the Satake-twisted classes of Theorem 1.5 — and the correct statement is the cocycle-range statement proved above. We record this as a correction of emphasis to [I], not of any equation.

1.4. Dissolution, and its exact purchase.

Corollary 1.7 (Dissolution of formulation). *M_∞ admits no faithful normal semifinite trace. Consequently no inequality of the form $\tau(x^*x) \geq 0$ with τ a trace is available on M_∞ , and the finite-level obstruction mechanism of [II, Thms. 1.5–1.6] — non-negativity of $\text{Tr}(\cdot^2)$ against $\text{Tr}(C^2) = -n(q-1) < 0$ — has no formulation on the weak closure of the tower. The obstruction is not overcome; the sentence expressing it ceases to exist.*

Proof. Type III means precisely the absence of a nonzero normal semifinite trace; the rest is the observation that the finite-level argument is a statement about a trace. $\square$

Proposition 1.8 (Inductive persistence). *Let $(\mathcal{E}_k)_{k \geq 0}$ be genuine C^* -correspondences over the level algebras A_k , compatible with the tower transfers, such that $[\mathcal{E}_k] \in KK^0(A_k, A_k)$ equals the level- k companion class C_k (equivalently, realizes $\Delta_{p,k}$ as its Pimsner index map). Then no such system exists: already each single level is impossible by [II, Thm. 1.6], and a fortiori no inductive limit $\mathcal{E}_\infty = \varinjlim \mathcal{E}_k$ realizes $[\mathcal{D}_p]_\infty$.*

Proof. Level by level, [II, Thm. 1.6]; there is nothing to add, which is the point. $\square$

Remark 1.9 (What dissolution buys, and what it destroys). Corollary 1.7 and Proposition 1.8 bracket the truth from two sides, and a third fact closes the bracket. The von Neumann completion that dissolves the trace also dissolves the prize: K_0 of a type III factor is trivial, so M_∞ retains no shadow of $\text{Jac}(Y_k)$ — the weak closure destroys exactly the invariant the programme exists to realize. The dissolution theorem therefore does *not* say “pass to the limit and construct”; it says that the unique *numerical* obstruction known dies at the measure-theoretic level, while the K -theoretic content survives only at the C^* -level. The sharpened open problem, which we state as the successor to [II, Conj. 3.6(iii)], is accordingly: does there exist a *traceless, purely infinite, nuclear* C^* -algebra over $A_\infty = C(\Omega)$ — a Kirchberg-type limit, where no trace exists to obstruct and where K -theory is nontrivial — carrying a genuine correspondence whose index map recovers the tower $(\mathbb{Z} \oplus \text{Jac}(Y_k))_k$? Tracelessness removes the only obstruction we have proved; it does not supply the construction, and we do not possess one.

1.5. The C^* -level: traceless with K -theory alive.

Proposition 1.10 (C^* -dissolution, no K -annihilation). *Let $C_r^*(\mathcal{G}_\infty)$ denote the reduced groupoid C^* -algebra of the rooted tail groupoid of [I, Def. 1.6] over Ω . It is a unital Kirchberg algebra: separable, nuclear, simple, purely infinite, and in the UCT class. Being purely infinite, it admits no nonzero trace, finite or semifinite, densely defined or bounded. Its K -theory is*

$$K_0(C_r^*(\mathcal{G}_\infty)) \cong \mathbb{Z}/(q-1), \quad K_1(C_r^*(\mathcal{G}_\infty)) = 0,$$

by stabilization to the Cuntz algebra $\mathcal{O}_q$ ([I, Prop. 1.7(iii)]). Consequently, at the C^ -level the trace obstruction of [II] has no formulation and K -theory is not annihilated: both properties denied to the von Neumann closure of Corollary 1.7 and Remark 1.9 hold simultaneously here.*

Remark 1.11 (What this algebra does not do, stated against temptation). It would be pleasant to add that $K_0(C_r^*(\mathcal{G}_\infty))$ “carries the tower of sandpile groups.” It does not, and asserting it would contradict this series’ own results: the boundary channel is arithmetically blind ([I, Thm. 3.7]), its K -theory seeing only $\mathbb{Z}/(q-1)$ locally and Euler-characteristic data globally, and nothing of $\text{Jac}(Y_k)$. Proposition 1.10 therefore proves an *existence of habitat*, not an occupancy: the C^* -level is the unique regime where tracelessness and nontrivial K -theory coexist, but the particular algebra naturally supplied by the tail groupoid is the wrong tenant. The sharpened open problem of Remark 1.9 stands exactly as posed: construct, functorially from the Hecke data, a Kirchberg algebra over $C(\Omega)$ whose K_0 realizes $(\mathbb{Z} \oplus \text{Jac}(Y_k))_k$; abstract existence for any prescribed UCT K -data is guaranteed by the Kirchberg–Phillips classification (quoted, data-unverified per the series policy), and canonicity, not existence, is the entire problem.

2. THE IWAHORI–HECKE EULER CHARACTERISTIC

The congruence (Iwahori path) tower of [II, §5] has levels $Y_k =$ non-backtracking k -paths of Y_0 , with $n_k = n(q+1)q^{k-1}$ vertices for $k \geq 1$, connected by the two degeneracy correspondences (initial and terminal truncation), and it is non-normal: no deck group acts, and [II, Rem. 5.2] showed its torsion growth has the exact pseudo-Iwasawa form with $\lambda = -1$, impossible for a $\mathbb{Z}_p[[T]]$ -characteristic ideal. This section derives that law from a single combinatorial input and locates the -1 .

Theorem 2.1 (Line-digraph recursion, proved). *Let G be a strongly connected digraph on n vertices in which every vertex has out-degree and in-degree $q \geq 2$ (Eulerian q -regular), $m = qn$ arcs, and let $L(G)$ be its line digraph. Then $L(G)$ is again strongly connected and Eulerian q -regular on m vertices, and*

$$|\text{sandpile}(L(G))| = q^{m-n-1} |\text{sandpile}(G)| = q^{n(q-1)-1} |\text{sandpile}(G)|.$$

In particular, along the Iwahori tower ($k \geq 1$), $|\text{sandpile}(Y_{k+1})| = q^{n_k(q-1)-1} |\text{sandpile}(Y_k)|$; for $q = 2$ this is the five-level-verified 2^{n_k-1} of [II, Rem. 5.2], whose exponent is a $q = 2$ coincidence, corrected here.

Proof. Step 1 (spectra). Let $D_{\text{out}}, D_{\text{in}} \in M_{n \times m}(\mathbb{Z})$ be the tail and head incidence matrices, $(D_{\text{out}})_{v,a} = [\text{tail}(a) = v]$, $(D_{\text{in}})_{v,a} = [\text{head}(a) = v]$. Then $A(G) = D_{\text{out}} D_{\text{in}}^t$ (arcs $u \rightarrow v$) and $A(L(G)) = D_{\text{in}}^t D_{\text{out}}$ (consecutive arc pairs). For rectangular $P \in M_{n \times m}$, $Q \in M_{m \times n}$ one has $\det(zI_m - QP) = z^{m-n} \det(zI_n - PQ)$; applied with $P = D_{\text{out}}$, $Q = D_{\text{in}}^t$,

$$(2.1) \quad \det(zI_m - A(L(G))) = z^{m-n} \det(zI_n - A(G)) :$$

the line digraph has the same nonzero spectrum with multiplicity, plus $m - n$ extra zeros. $L(G)$ is Eulerian q -regular because both degrees of an arc equal the corresponding degrees of its endpoints in G , and strongly connected because a path between two arcs is obtained by concatenating a vertex path of G from the head of the first to the tail of the second.

Step 2 (equal cofactors and the eigenvalue product). Let $\Delta = qI - A$ for an Eulerian q -regular strongly connected digraph on N vertices. Row sums of Δ vanish (out-degree q) and column sums vanish (in-degree q), so the all-ones vector spans both the right and the left kernel; by Perron–Frobenius, q is an algebraically simple eigenvalue of the irreducible non-negative matrix A , so 0 is a simple eigenvalue of Δ . The two-sided kernel $\mathbb{C}\mathbf{1}$ forces all N principal cofactors of Δ to be equal — cofactors agree along a row because the columns of Δ sum to zero, and along a column because the rows do — and the common value is by definition $\kappa := |\text{sandpile}|$, the determinant of the reduced Laplacian at any sink; sink-independence is exactly this equality. Expanding $\det(zI - \Delta) = z \prod_{i=2}^N (z - \mu_i)$ over the nonzero eigenvalues μ_i , the coefficient of z equals $(-1)^{N-1} \prod_{i \geq 2} \mu_i$ and also $(-1)^{N-1} \sum(\text{principal cofactors}) = (-1)^{N-1} N\kappa$, whence

$$(2.2) \quad \prod_{\mu \in \text{Spec } \Delta \setminus \{0\}} \mu = N\kappa.$$

Step 3 (assembly). Apply (2.2) to G and to $L(G)$. By (2.1), the nonzero eigenvalues of $\Delta_{L(G)} = qI_m - A(L(G))$ are those of Δ_G together with $q - 0 = q$ taken with multiplicity $m - n$; the eigenvalue 0 remains simple for $\Delta_{L(G)}$ since q is Perron-simple for $A(L(G))$ as well. Hence

$$m\kappa(L(G)) = q^{m-n} \cdot n\kappa(G) \implies \kappa(L(G)) = \frac{n}{m} q^{m-n} \kappa(G) = q^{m-n-1} \kappa(G),$$

using $m = qn$. Besides the five $q = 2$ tower levels, the formula was confirmed exactly in a $q = 3$ instance (the complete digraph on four vertices: $\kappa = 16$, $\kappa(L) = 34992 = 3^7 \cdot 16$), against which the uncorrected exponent $n - 1$ fails by a factor 3^4 . $\square$

Theorem 2.2 (The pseudo-Iwasawa law). *Let the base graph have n vertices and $q = p$. Then, unconditionally by Theorem 2.1, for $k \geq 2$,*

$$\text{ord}_p |\text{sandpile}(Y_k)| = \underbrace{\frac{n(q+1)}{q}}_{\mu} p^k - k + \nu_0, \quad \lambda = -1,$$

with ν_0 determined by the first level. For $Y_0 = K_4$, $q = 2$: $\mu = 6$, $\nu_0 = -4$, reproducing $\nu_2 = 6 \cdot 2^k - k - 4$ exactly.

Proof. Summing the recursion of Theorem 2.1 from level 1,

$$\text{ord}_p |\text{sandpile}(Y_k)| = \text{ord}_p |\text{sandpile}(Y_1)| + \sum_{j=1}^{k-1} (n_j(q-1) - 1),$$

the multiplier being a pure power of $p = q$, and $\sum_{j=1}^{k-1} n_j(q-1) = n(q+1)(q^{k-1}-1) = \frac{n(q+1)}{q} p^k - n(q+1)$. With $q = 2$, $n = 4$: $\sum_{j=1}^{k-1} (12 \cdot 2^{j-1} - 1) = 6 \cdot 2^k - 12 - (k-1)$, and $\text{ord}_2 |\text{sandpile}(Y_1)| = 7$ gives $6 \cdot 2^k - k - 4$. The general coefficients read off the same computation. $\square$

Proposition 2.3 ($\lambda \geq 0$ for commutative Iwasawa towers). *Let $g \in \mathbb{Z}_p[[T]]$ be nonzero with Weierstrass decomposition $g = p^\mu P(T)U(T)$, P distinguished of degree $\lambda \geq 0$, U a unit. Then for $k \gg 0$, $\text{ord}_p \prod_{\zeta^{p^k}=1, \zeta \neq 1} g(\zeta-1) = \mu(p^k-1) + \lambda k + O(1)$ with $\lambda \geq 0$. Hence no tower governed by the characteristic ideal of a finitely generated torsion $\mathbb{Z}_p[[T]]$ -module can exhibit $\lambda = -1$: Theorem 2.2 is a rigorous diagnostic that the Iwahori tower is not such a tower.*

Proof. $\text{ord}_p g(\zeta_{p^j} - 1) = \mu + \lambda \text{ord}_p(\zeta_{p^j} - 1) + O(p^{-j}) = \mu + \lambda/\varphi(p^j) + o(1)$ for $j \gg 0$ by the decomposition, and summing over the $\varphi(p^j)$ primitive roots for $1 \leq j \leq k$ gives the formula; λ is a degree, hence non-negative. $\square$

Definition 2.4 (Degeneracy pair and its Euler term). Let $s_k, t_k: A_{k+1} \rightrightarrows A_k$ be the initial- and terminal-truncation correspondences of the tower, the graph-level avatars of the two degeneracy maps $X_0(p^{k+1}) \rightrightarrows X_0(p^k)$, generating the action of the Iwahori–Hecke algebra $\mathcal{H}(\text{PGL}_2(\mathbb{Q}_p), \mathcal{I})$ on the tower. The *Eisenstein kernel* at level k is the one-dimensional space of $\mathcal{R}$ -harmonic constants (the Perron direction), simultaneously the kernel of the level- k sandpile Laplacian and the sole obstruction to exactness of the pair (s_k, t_k) on 0-chains.

Theorem 2.5 (The -1 is an Euler characteristic). *The transfer of torsion along one level of the Iwahori tower factors, by Theorem 2.1, as*

$$\text{ord}_p |\text{sandpile}(Y_{k+1})| - \text{ord}_p |\text{sandpile}(Y_k)| = \underbrace{n_k(q-1)}_{\text{arc growth: } \text{ord}_p q^{m_k - n_k}} - \underbrace{\chi_k}_{=1},$$

where $\chi_k = \dim \ker \Delta_k^{\text{dir}} = 1$ is the rank of the Eisenstein kernel removed at each level. Hence in the growth law of Theorem 2.2 the invariants separate cleanly: μ is the arc-proliferation density $n(q+1)/q$ of the correspondences, and $\lambda = -\chi = -1$ is the Euler-characteristic correction of the degeneracy pair — a homological constant of the non-commutative base $\mathcal{H}(\text{PGL}_2(\mathbb{Q}_p), \mathcal{I})$, not a module invariant over any commutative $\mathbb{Z}_p[[T]]$. In particular the corrected form of [II, Conj. 3.6(ii-b)] should assert the following: the characteristic object of the congruence tower is a module over $\mathcal{H}(\text{PGL}_2(\mathbb{Q}_p), \mathcal{I})$ whose growth law carries the intrinsic correction $-\chi \cdot k$, and it reduces to a commutative characteristic ideal exactly on sub-towers with abelian deck action, where the χ -correction vanishes and the Newton identities of [II, §5] hold to the integer.

Proof. The recursion's exponent splits as $(m_k - n_k) - 1 = \text{ord}_p(q^{m_k - n_k}) - 1$, and both terms are now located inside the proof of Theorem 2.1 itself. The first is the p -valuation of the extra Perron factors: each of the $m_k - n_k$ new zero eigenvalues of $A(Y_{k+1})$ contributes $q - 0 = q$ to the eigenvalue product (2.2). The subtracted 1 is the vertex-count normalization $n_k/m_k = q^{-1}$ in (2.2): precisely the one-dimensional Eisenstein kernel $\ker \Delta^{\text{dir}}$, simple at every level by strong connectivity, excluded from the eigenvalue product once per level, independently of k . Summation as in Theorem 2.2 turns the per-level $-\chi_k = -1$ into the global $-k$, i.e. $\lambda = -\chi = -1$. $\square$

Remark 2.6 (Honest scope). Theorem 2.5 is now unconditional, but its final sentence remains a proposal for restating [II, Conj. 3.6(ii-b)], not a theorem about $\mathcal{H}(\text{PGL}_2(\mathbb{Q}_p), \mathcal{I})$ -modules: we have defined no characteristic ideal over the Iwahori–Hecke algebra, and constructing that module theory — with a Weierstrass-type preparation in which $-\chi$ appears intrinsically — is the open algebraic problem this section poses. What is unconditional is Proposition 2.3: $\lambda = -1$ certifies, rigorously, that whatever that theory is, it is not commutative Iwasawa theory.

3. EXPLORATORY OUTLOOK

We record, deriving nothing, where the two threads point. The dissolution theorem directs the realization problem toward traceless Kirchberg limits over $C(\Omega)$ (Remark 1.9), where the natural analytic invariants are not traces but the KMS functionals of the twisted dynamics of [I, §2], with local L -factors as partition functions. The Euler-characteristic correction of Section 2 lives at the trivial zero of those L -factors — the same Eisenstein direction removed once per level — and it is natural to ask, in the spirit of the exceptional-zero phenomena of Mazur–Tate–Teitelbaum and of Gross–Stark, whether the per-level derivative data of the tower assembles into a p -adic L -functional on the Iwahori–Hecke base whose trivial-zero corrections reproduce $-\chi k$. We state this as a direction, not a conjecture: no formulation precise enough to be false is offered here, and the series’ methodology is to compute before naming. The first computable instance has in fact been computed. Write $g(T) = T^2 h(T)$ for the cleared voltage pencil (the double trivial zero of [II, Rem. 5.4]) and $\varepsilon_j := \text{ord}_\ell \prod_{\zeta \text{ prim. } \ell^j} h(\zeta - 1) - (\mu \varphi(\ell^j) + \lambda_h)$ for the exceptional low-level defects. Then the constant term of the growth law satisfies, exactly in all four abelian towers of [II, Table 2],

$$\nu_0 = \text{ord}_\ell \kappa(Y_0) - \mu + \sum_{j \geq 1} \varepsilon_j,$$

with ε finitely supported: identically zero in three towers, and $(\varepsilon_1, \varepsilon_2) = (-1, +2)$ in the $(1, 2, 0)$ tower. The trivial-zero residue $a_2 = \frac{1}{2}g''(0)$ carries the leading datum precisely when $\lambda_h = 0$, where $\text{ord}_\ell a_2 = \mu$ as in both $(1, 0, 0)$ towers, and not otherwise: an exceptional-zero reading in the Mazur–Tate–Teitelbaum style is available only in the $\lambda_h = 0$ regime, and this bookkeeping identity, verified to the integer, is the whole of what is presently proved.

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Unrefereed preprint. Proved in full: Lemmas 1.2–1.3, Theorem 1.4, Corollary 1.7, Propositions 1.8 and 2.3, Theorem 2.1 (self-contained; the only classical input is the algebraic simplicity of the Perron eigenvalue of an irreducible non-negative matrix), Theorem 2.2, and Theorem 2.5, all now unconditional. Quoted classical inputs: every entry in the bibliography is verified against its primary source or Crossref record — Connes and Powers via JSTOR, Araki–Woods via EMS Press, Krieger via Springer, Feldman–Moore Parts I and II via Crossref and the AMS Digital Library, Levine via Crossref — with no residual soft spots; the DOIs are recorded inline at each entry. The Levine reference [6] is retained for priority context only: Theorem 2.1 is proved self-containedly above, and comparing the exponent $q^{n(q-1)-1}$ with the published statement belongs to the primary-source check this series requires. The concluding restatement of [II, Conj. 3.6(ii-b)] in Theorem 2.5 remains a proposal. Remark 1.6 records a correction of emphasis to [I, Thm. 2.4(iii)]. Section 3 makes no claims.

REFERENCES

- [1] R. T. Powers, *Representations of uniformly hyperfinite algebras and their associated von Neumann rings*, Ann. of Math. (2) **86** (1967), no. 1, 138–171. [Primary-source verified: JSTOR, DOI 10.2307/1970364.]
- [2] H. Araki and E. J. Woods, *A classification of factors*, Publ. Res. Inst. Math. Sci. **4** (1968), no. 1, 51–130. [Primary-source verified: EMS Press / PRIMs article page.]
- [3] W. Krieger, *On ergodic flows and the isomorphism of factors*, Math. Ann. **223** (1976), 19–70. [Primary-source verified: Springer, Math. Ann. **223**, pp. 19–70, February 1976.]
- [4] A. Connes, *Classification of injective factors. Cases II_1 , II_∞ , III_λ , $\lambda \neq 1$* , Ann. of Math. (2) **104** (1976), no. 1, 73–115. [Primary-source verified: JSTOR, DOI 10.2307/1971057.]
- [5] J. Feldman and C. C. Moore, *Ergodic equivalence relations, cohomology, and von Neumann algebras. I, II*, Trans. Amer. Math. Soc. **234** (1977), 289–324 and 325–359. [Both parts verified: Part I Crossref-verified, DOI 10.1090/S0002-9947-1977-0578656-4, pp. 289–324; Part II primary-source verified, DOI 10.1090/S0002-9947-1977-0578730-2, pp. 325–359. The related 1975 Bulletin announcement, Bull. Amer. Math. Soc. **81** (1975), no. 5, 921–924, DOI 10.1090/S0002-9904-1975-13888-2, is a distinct paper and is not the reference intended here.]

- [6] L. Levine, *Sandpile groups and spanning trees of directed line graphs*, J. Combin. Theory Ser. A **118** (2011), no. 2, 350–364. [Crossref-verified: DOI 10.1016/j.jcta.2010.04.001.]

GUT GEOSERVICE INC., RIVIÈRE-BEAUDETTE, QUÉBEC, CANADA

Email address: `ruqing@hotmail.com`